**Akalanka Galappaththi, PhD**
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## Professional Summary

Data Scientist with 7+ years of experience applying statistical modeling, machine learning, and NLP to large-scale, complex, and unstructured datasets. PhD-trained in rigorous experimental methodology with a track record of delivering measurable performance improvements on ambiguous, open-ended problems – including building production-grade ML systems from model validation through REST API deployment. Experienced collaborating cross-functionally with engineering and product teams, mentoring junior contributors, and translating complex technical findings into clear decisions for non-technical stakeholders. Eager to apply this foundation to cybersecurity applications where data-driven models have high real-world impact.

## Technical Skills

- **Machine Learning & Statistics:** Statistical modeling, regression, clustering, hypothesis testing, A/B testing, power analysis, model evaluation (precision, recall, F1, pass@k), NLP (TF-IDF, word embeddings), LLMs & Generative AI (prompt engineering, RAG, function calling), feature engineering
- **Frameworks & libraries:** Python (pandas, NumPy, scikit-learn, PyTorch), REST APIs (FastAPI), web scraping (beautifulsoup, playwright)
- **Data Engineering & infrastructure:** PostgreSQL, MySQL, Firebase, data modeling, SQL (CTEs, window functions, complex joins), ETL pipeline development, Docker
- **Visualization & Communication:** Tableau, Power BI, Matplotlib, Plotly, notebooks
- **Software Engineering practices:** Git/GitHub, Agile/Scrum, documentation

## Projects

**Credit Risk Decision System – End-to-End ML Service** · Solo project · In progress · 2026
Production-grade ML system applying rigorous statistical methodology to real-world credit adjudication – from raw data through containerized REST API deployment · https://github.com/boneyag/CreditRisk

- Designed and validated a challenger model (XGBoost vs. logistic regression baseline) using rigorous hypothesis testing: McNemar's test yielded $p = 4.1e-38$ and bootstrap CIs on accuracy delta confirmed a statistically robust +3.7% gain.
- Engineered a modular, containerised training pipeline (Docker Compose) with decoupled training/serving stages, versioned artifact manifests, and structured JSON audit logging on every prediction – mirroring production ML deployment practices.
- Built a FastAPI REST service with automated endpoint tests (pytest) and Pydantic schema validation, achieving environment parity between training and inference containers.

- Optimised model for risk-centric recall on high-default applicants (F1-score: 0.92), translating business objectives into metric selection decisions rather than defaulting to accuracy – consistent with real credit risk constraints.
- Extending system with SHAP-based explanations per prediction for adjudicator transparency, and AWS Fargate deployment with Prometheus monitoring for real-time feature drift detection.

## Professional Experience

**Data Scientist – Applied ML & Experimentation** · University of Alberta · 2020–2025

- Designed and built scalable ETL pipelines using Python and REST APIs to collect and process large-scale structured and unstructured data (GitHub repositories, Stack Overflow posts), enabling multi-year analytical programs across research and industry partnerships.
- Developed LLM-based solutions for automated code analysis, improving bug detection accuracy by 22% and fix effectiveness by 63% over baseline – reducing the manual debugging cycle developers face with data science library errors.
- Conducted exploratory data analysis on high-dimensional datasets to uncover behavioral patterns and inform model design and prompt strategies.
- Designed and executed controlled experiments using parametric and non-parametric statistical tests; managed sample size decisions, confidence levels, and error margin trade-offs to ensure reliable, reproducible results.
- Defined and tracked core performance metrics (precision, recall, F1-score, pass@k), creating a single source of truth for model evaluation across research leads and industry stakeholders.
- Built modular, well-documented data workflows in Python and SQL, maintaining integrity across the full analytical lifecycle from raw collection through reporting.
- Communicated complex analytical findings through written reports and presentations to technical and non-technical audiences, driving alignment under ambiguous and evolving requirements.

**Data Analytics Consultant** · Triggericon · 2024–2025

- Designed and executed A/B experiments to evaluate the relative impact of Google Ads vs. social media campaigns – including hypothesis framing, alpha/beta calibration, and sample size planning – delivering a statistically grounded channel recommendation to stakeholders.
- Built a reusable Python sample size planning tool, enabling the team to independently scope future experiments without analyst involvement for routine campaign decisions.
- Analyzed consumer engagement data from Google Analytics across click-through rates, cost-per-click, and conversion rates, identifying optimization opportunities that informed campaign budget adjustments.
- Developed an executive revenue dashboard tracking total revenue, month-over-month growth, cost-vs-revenue by channel, and new vs. returning customer segmentation.

**Technical Project Lead & Coordinator** · University of Alberta · 2020–2025

- Mentored 20+ cross-functional project teams, defining requirements and delivering technical solutions under real-world constraints.
- Managed full project lifecycles using Agile (Scrum/Kanban), tracking individual and team performance to ensure timely delivery.
- Guided teams through full software engineering lifecycle including requirements gathering, UML modeling, testing strategies, and technical documentation under real-world project constraints.

**Data Analyst – Research & NLP** · University of Lethbridge · 2018–2020

- Developed machine learning models for automated text analysis and summarization, achieving a 23% performance improvement over industry baseline through iterative feature engineering and statistical validation.
- Collected, cleaned, and validated large-scale text datasets; applied NLP techniques (TF-IDF, word embeddings) to extract structured signals from unstructured sources.
- Evaluated model performance using statistical metrics and communicated findings through visualizations and technical reports to guide project decision-making.

## Education

- PhD, Computing Science (Software Engineering) – University of Alberta, 2025
- MSc, Computer Science – University of Lethbridge, 2020
- MPhil, Computer Science – University of Peradeniya, Sri Lanka, 2018
- BSc, Computer Science – University of Peradeniya, Sri Lanka, 2010

## Recognition

- Competitive scholarships: Government of Alberta & University of Alberta – CAD 55K+ total, awarded for research excellence
- Program Committee Member: international peer-reviewed conferences – evaluated experimental design and analytical methodology